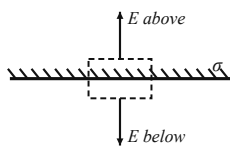


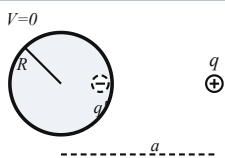
1 · Set up first (most of the points)

- Write: region, coordinates, **numbered BCs**, general solution. Then solve.
- Grounded: $V = 0$. Held at V_0 : $V = V_0$. Isolated + neutral: $V = \text{unknown const}$, total charge 0. Far away: $V \rightarrow 0$ (uniform field: $V \rightarrow -E_0 r \cos \theta$).
- Always: V finite at the center; V continuous everywhere.
- Across a charged surface: $E_\perp$ jumps by σ/ϵ_0 . Same thing: $\partial_n V_{out} - \partial_n V_{in} = -\sigma/\epsilon_0$.
- On metal: $\sigma = -\epsilon_0 \partial V / \partial n$. Field just outside = σ/ϵ_0 , straight out.



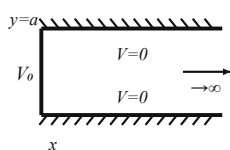
2 · Method of images

- **Plane**, q at height d : image $-q$ at depth d . Force $\frac{kq^2}{(2d)^2}$ toward plane. Energy $-\frac{kq^2}{4d}$ (half the pair).
- $\sigma = -\frac{qd}{2\pi(s^2+d^2)^{3/2}}$, total $-q$.
- **Grounded sphere** R , q at distance a : image $q' = -\frac{qR}{a}$ at $\frac{R^2}{a}$ from center. Force $\frac{kq^2 Ra}{(a^2 - R^2)^2}$ (pull).
- Twist = extra charge at the **center**: held at $V_0 \rightarrow 4\pi\epsilon_0 R V_0$; neutral $\rightarrow +\frac{qR}{a}$; charge $Q \rightarrow Q + \frac{qR}{a}$.
- q **inside** grounded shell at d : image $-\frac{qR}{d}$ at $\frac{R^2}{d}$ (outside).
- 90° corner: 3 images $(-q, -q, +q)$. Line charge: image $- \lambda$.
- Only valid outside the metal. Check every BC $\rightarrow$ uniqueness says it's right.



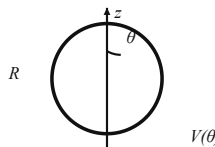
3 · Separation: boxes (Cartesian)

- $V = X(x)Y(y)$. Direction with **two zero walls** gets $\sin \frac{n\pi y}{a}$.
- Other direction: $e^{-n\pi x/a}$ (open to ∞), $\sinh \frac{n\pi(b-x)}{a}$ (zero at $x = b$), cosh (symmetric).
- $C_n = \frac{2}{a} \int_0^a V_0(y) \sin \frac{n\pi y}{a} dy$. Constant V_0 : $C_n = \frac{4V_0}{n\pi}$, odd n only.
- Live face at $x = b$: divide by $\sinh \frac{n\pi b}{a}$. Two live faces: solve each, add.
- At a jump in the edge value, the series gives the average.



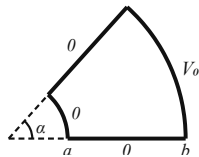
4 · Separation: spheres (Legendre)

- $V = \sum (A_\ell r^\ell + \frac{B_\ell}{r^{\ell+1}}) P_\ell(\cos \theta)$.
- Inside: only $A_\ell r^\ell$. Outside: only $\frac{B_\ell}{r^{\ell+1}}$. **Not $r^{-\ell}$** .
- $x = \cos \theta$: $P_0 = 1$, $P_1 = x$, $P_2 = \frac{3x^2-1}{2}$, $P_3 = \frac{5x^3-3x}{2}$. Every $P_\ell(1) = 1$.
- $\cos^2 \theta = \frac{1}{3} + \frac{2}{3} P_2$. $\cos^3 \theta = \frac{3}{5} P_1 + \frac{2}{5} P_3$.
- Anything else: $c_\ell = \frac{2\ell+1}{2} \int_{-1}^1 f(x) P_\ell(x) dx$.
- Shell at $V(\theta) = \sum c_\ell P_\ell$: inside $c_\ell (\frac{r}{R})^\ell P_\ell$, outside $c_\ell (\frac{R}{r})^{\ell+1} P_\ell$, $\sigma = \frac{\epsilon_0}{R} \sum (2\ell+1) c_\ell P_\ell$.
- Shell with $\sigma = \sum \sigma_\ell P_\ell$: inside $A_\ell = \frac{\sigma_\ell}{(2\ell+1)\epsilon_0 R^{\ell-1}}$, outside $B_\ell = A_\ell R^{2\ell+1}$.
- Metal sphere in uniform E_0 (up): $V = -E_0(r - \frac{R^3}{r^2}) \cos \theta$, $\sigma = 3\epsilon_0 E_0 \cos \theta$.



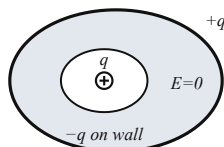
5 · Separation: wedges and pipes (polar)

- Long in z . $\Theta = \sin k\theta$, $\cos k\theta$; $R = r^k$, r^{-k} ($k = 0$: $\ln r$, θ).
- Walls at $\theta = 0, \alpha$ grounded: $\sin \frac{n\pi\theta}{\alpha}$, $k = \frac{n\pi}{\alpha}$.
- Has the center: only r^k . Goes to ∞ : only r^{-k} . Zero arc at a : $(\frac{r}{a})^k - (\frac{a}{r})^k$.
- Full circle: $k = 1, 2, 3, \dots$. One wall at V_0 , other 0, no arcs: $V = V_0 \theta / \alpha$.



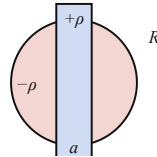
6 · Conductors and capacitors

- Inside metal: $E = 0$, no charge, one value of V . Charge sits on the surface.
- Empty cavity: $E = 0$ inside it. Charge q in cavity: $-q$ on the cavity wall, $+q$ (+ net Q) outside.
- $C = Q/\Delta V$. Plates $\frac{\epsilon_0 A}{d}$. Spheres $\frac{4\pi\epsilon_0 ab}{b-a}$. One sphere $4\pi\epsilon_0 R$. Coax $\frac{2\pi\epsilon_0 L}{\ln(b/a)}$. Energy $\frac{1}{2} CV^2$.



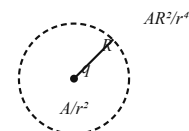
7 · Gauss fields (magnitudes, outward)

- Outside a ball or point: $\frac{kQ}{r^2}$. Inside uniform ball: $\frac{\rho r}{3\epsilon_0}$. Inside a shell: 0.
- Line: $\frac{\lambda}{2\pi\epsilon_0}$. Inside uniform cylinder: $\frac{\rho s}{2\epsilon_0}$.
- Sheet: $\frac{\sigma}{2\epsilon_0}$ each side. Metal surface: $\frac{\sigma}{\epsilon_0}$.
- Hole = full shape + a negative piece. Each field from **its own** center, then add.
- Ball potential inside: $V = \frac{kQ(3R^2 - r^2)}{2R^3}$.



8 · Given E, find rho (delta at the origin)

- Away from 0: $\rho = \frac{\epsilon_0}{r^2} \frac{d}{dr}(r^2 E)$.
- Charge inside radius r : $4\pi\epsilon_0 r^2 E$. If it tends to $q \neq 0$ as $r \rightarrow 0$: point charge $q \delta^3$ at the origin.
- E jumps at R : $\sigma = \epsilon_0(E_{out} - E_{in})$. Total: $4\pi\epsilon_0 r^2 E$ as $r \rightarrow \infty$.
- $\nabla^2 \frac{1}{r} = -4\pi\delta^3$. $\int f(x)\delta(x-a)dx = f(a)$.



9 · Energy

- Charges: $W = \frac{1}{2} \sum q_i V_i$ (V from the others). Spread out: $W = \frac{1}{2} \int \rho V = \frac{\epsilon_0}{2} \int E^2$.
- Ball $\frac{3}{5} \frac{kQ^2}{R}$. Shell $\frac{1}{2} \frac{kQ^2}{R}$. Pair $\frac{kq_1 q_2}{r}$. Not additive.

10 · Laplace and uniqueness

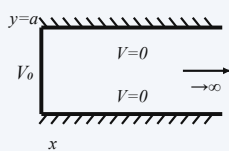
- No charge: $\nabla^2 V = 0$. No max or min inside; $V = \text{average over a sphere around the point}$.
- V given on every boundary (or Q on each metal) $\rightarrow$ only one answer. Anything that fits is right.

11 · Traps + integrals

- $d\tau = r^2 \sin \theta dr d\theta d\phi$. **Metal σ/ϵ_0 , sheet $\sigma/2\epsilon_0$. Use $\cos \theta$ inside P_ℓ .**
- $\int \frac{dx}{(x^2+a^2)^{3/2}} = \frac{x}{a^2 \sqrt{x^2+a^2}}$. $\int \frac{x dx}{(x^2+a^2)^{3/2}} = \frac{-1}{\sqrt{x^2+a^2}}$.
- $\int_0^a \sin^2 \frac{n\pi y}{a} dy = \frac{a}{2}$. $\int_0^\pi \sin^3 \theta d\theta = \frac{4}{3}$.

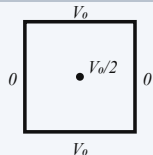
Ex 1 · Slot (Cartesian)

- BCs: (1) $V = 0$ at $y = 0$; (2) $V = 0$ at $y = a$; (3) $V \rightarrow 0$ as $x \rightarrow \infty$; (4) $V = V_0$ at $x = 0$.
- (1)(2) $\rightarrow \sin \frac{n\pi y}{a}$. (3) $\rightarrow e^{-n\pi x/a}$. (4) $\rightarrow C_n = \frac{4V_0}{n\pi}$, odd n .
- $V = \frac{4V_0}{\pi} \sum_{\text{odd } n} \frac{1}{n} e^{-n\pi x/a} \sin \frac{n\pi y}{a}$.
- Closed at $x = b$: use $\frac{\sinh(n\pi(b-x)/a)}{\sinh(n\pi b/a)}$ instead.



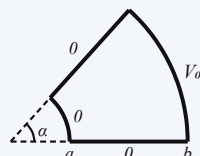
Ex 2 · Box, top and bottom at V_0

- Sides zero $\rightarrow \sin \frac{n\pi x}{a}$. Symmetric top/bottom $\rightarrow \cosh \frac{n\pi(y-a/2)}{a}$.
- $C_n \cosh \frac{n\pi}{2} = \frac{4V_0}{n\pi}$, odd n .
- Center: $V_0/2$ (this + the turned-sideways problem = every face at V_0).



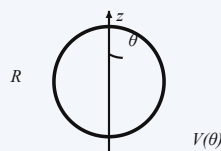
Ex 3 · Wedge (Spring '26 exam)

- $a < r < b$. BCs: $V = 0$ on both walls and on $r = a$; $V = V_0$ on $r = b$.
- Walls $\rightarrow \sin \frac{n\pi\theta}{\alpha}$, $k = \frac{n\pi}{\alpha}$. Zero at $a \rightarrow (\frac{r}{a})^k - (\frac{a}{r})^k$.
- $C_n = \frac{4V_0}{n\pi[(b/a)^k - (a/b)^k]}$, odd n .



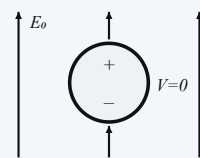
Ex 4 · Shell at $V_0 \cos^2 \theta$

- $\cos^2 \theta = \frac{1}{3} + \frac{2}{3} P_2$.
- Inside: $V_0 [\frac{1}{3} + \frac{2}{3} \frac{r^2}{R^2} P_2]$.
- Outside: $V_0 [\frac{R}{3r} + \frac{2}{3} \frac{R^3}{r^3} P_2]$.
- $\sigma = \frac{\epsilon_0 V_0}{R} [\frac{1}{3} + \frac{10}{3} P_2]$.



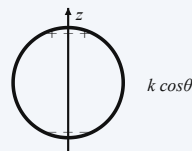
Ex 5 · Metal sphere in a uniform field

- BCs: $V = 0$ on sphere; far away $V \rightarrow -E_0 r \cos \theta$.
- Only $\ell = 1$: $V = -E_0 r \cos \theta + \frac{B \cos \theta}{r^2}$. $V(R) = 0 \rightarrow B = E_0 R^3$.
- $\sigma = 3\epsilon_0 E_0 \cos \theta$: + on top, - on bottom.



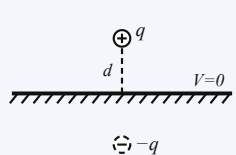
Ex 6 · Shell with $\sigma = k \cos \theta$

- Inside $A r \cos \theta$; outside $\frac{A R^3 \cos \theta}{r^2}$ (matches at R).
- Jump: $-2A - A = -\frac{k}{\epsilon_0} \rightarrow A = \frac{k}{3\epsilon_0}$.
- Inside: uniform field $\frac{k}{3\epsilon_0}$ pointing down.



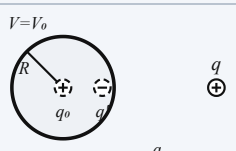
Ex 7 · Charge over a grounded plane

- $V = kq [\frac{1}{\text{dist to } q} - \frac{1}{\text{dist to image}}]$.
- $\sigma = -\frac{q d}{2\pi(s^2 + d^2)^{3/2}}$, total $-q$; half lies within $s = \sqrt{3}d$.
- Force $\frac{kq^2}{4d^2}$ down. Energy $-\frac{kq^2}{4d}$.



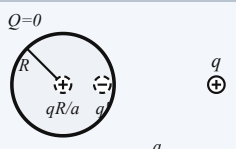
Ex 8 · Sphere held at V_0 , charge q at a

- Images: $q' = -\frac{qR}{a}$ at $\frac{R^2}{a}$, plus $q_0 = 4\pi\epsilon_0 R V_0$ at center.
- Sphere's charge: $q_0 - \frac{qR}{a}$.
- Force on q : $\frac{qR V_0}{a^2}$ (push) $-\frac{kq^2 R a}{(a^2 - R^2)^2}$ (pull).
- Zero at $a = 2R$ when $V_0 = \frac{8}{9} \frac{kq}{R}$. Unstable.



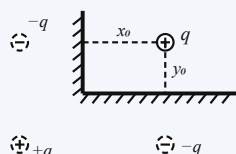
Ex 9 · Neutral sphere, charge q at a

- Images: $-\frac{qR}{a}$ at $\frac{R^2}{a}$ and $+\frac{qR}{a}$ at center.
- Always pulls q in. Sphere's $V = \frac{kq}{a}$.



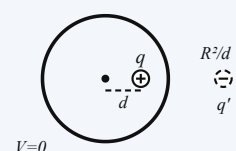
Ex 10 · Charge in a 90° corner

- q at (x_0, y_0) . Images: $-q$ at $(-x_0, y_0)$, $-q$ at $(x_0, -y_0)$, $+q$ at $(-x_0, -y_0)$.
- Force: add the three pulls/pushes as vectors.



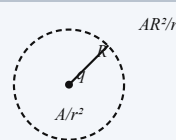
Ex 11 · Charge inside a grounded shell

- Image $-\frac{qR}{d}$ at $\frac{R^2}{d}$ (outside).
- Force $\frac{kq^2 R d}{(R^2 - d^2)^2}$ toward the near wall. Inner wall holds $-q$.



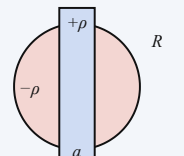
Ex 12 · Given E , find ρ and V

- $E = \frac{A}{r^2}$ inside R , $\frac{A R^2}{r^4}$ outside, outward.
- Inside: $r^2 E = A$ constant $\rightarrow$ no ρ for $r > 0$; point charge $4\pi\epsilon_0 A$ at origin.
- Outside: $\rho = -\frac{2\epsilon_0 A R^2}{r^5}$. E continuous at $R \rightarrow \sigma = 0$. Total 0.
- $V_{\text{out}} = \frac{A R^2}{3r^3}$, $V_{\text{in}} = \frac{A}{r} - \frac{2A}{3R}$.



Ex 13 · Cylinder through a ball

- Cylinder $+\rho$ (radius a , along z), ball $-\rho$ (radius R).
- Inside both: sideways $\frac{\rho s}{6\epsilon_0}$ out, along z : $\frac{\rho z}{3\epsilon_0}$ toward center.
- Plane $z = 0$, $a < s < R$: $\frac{\rho a^2}{2\epsilon_0 s} - \frac{\rho s}{3\epsilon_0}$, zero at $s = 1.22a$.



Ex 14 · Energy of a uniform ball

- Inside $E = \frac{kQr}{R^3}$, outside $\frac{kQ}{r^2}$.
- $\frac{\epsilon_0}{2} \int E^2 4\pi r^2 dr = \frac{3}{5} \frac{kQ^2}{R}$.

